PSet 2

1) C

2) a) $\begin{bmatrix} 9 \\ 12 \\ 3 \end{bmatrix} \begin{matrix} \rightarrow wx \\ \rightarrow wy \\ \rightarrow w \end{matrix}$

$\downarrow$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{9}{3} \\ \frac{12}{3} \end{bmatrix} \boxed{= (3, 4)}$

b) $\begin{bmatrix} 9 \\ 12 \\ 3 \end{bmatrix} \boxed{= (9, 12, 3, 1)}$

append a $\underline{1}$

3) A

4) a) x-axis

b) B

c) B

d) C

e) C

5) a) Extrinsic matrix: $E = [R|t]$

camera axis	world axis:
X_c	x_w
Y_c	Z_w
Z_c	$-Y_w$

camera-real world offset:

$C = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}$

$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

$t = -RC = -\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix} = -\begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ -5 \end{bmatrix}$

$\boxed{E = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & -1 & 0 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right]}$

b) ${}^{c}p = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} \cdot {}^{w}p$

$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & -1 & 0 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 5 \\ 3 \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} -1 \\ -1 \\ -10 \\ 1 \end{bmatrix}}$

c) $\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underset{\text{intrinsic matrix}}{K} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \underbrace{\underset{\text{extrinsic matrix}}{[R|t]} \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}}$ $\boxed{(u, v) = (210, 110)}$

$= \begin{bmatrix} 100 & 0 & 200 \\ 0 & 100 & 100 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} -2100 \\ -1100 \\ -10 \end{bmatrix} = -10 \begin{bmatrix} 210 \\ 110 \\ 1 \end{bmatrix}$

d) Projective Signed dist. $= \frac{z_c}{z_{max}} = \frac{-1}{2.8}$

$\boxed{\approx -3.57}$

6) a) c

b) b

c) c

d) b

e) $z = 5.2,\ \sigma = 1$

A: Likelihood $= e^{\left(-\frac{(5.2-4)^2}{2}\right)} \approx 0.49$

B: Likelihood $= e^{\left(-\frac{(5.2-5)^2}{2}\right)} \approx 0.98$

C: Likelihood $= e^{\left(-\frac{(5.2-6)^2}{2}\right)} \approx 0.73$

Unnormalized weights:

A: $0.3 \cdot 0.49 \approx 0.147$

B: $0.4 \cdot 0.98 \approx 0.392$

C: $0.3 \cdot 0.73 \approx 0.219$

Normalized:

$\sum$ unnormalized weights $= 0.758$

$$\boxed{\begin{aligned} A&: \frac{0.147}{0.758} \approx 0.194 \\ B&: \frac{0.392}{0.758} \approx 0.517 \\ C&: \frac{0.219}{0.758} \approx 0.289 \end{aligned}}$$